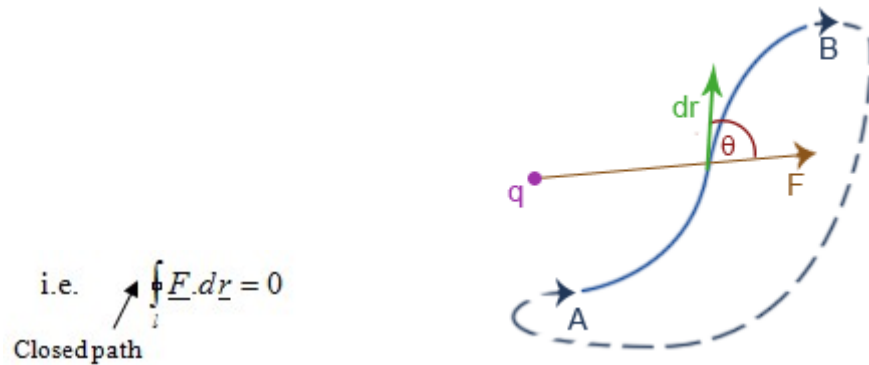


Potential energy

Electric force is conservative force

Electric field is a conservative field.

Conservative field – A field for which the work done in moving an isolated particle of unit mass/charge around a closed path is zero.



Where, F = the force acting on the particle

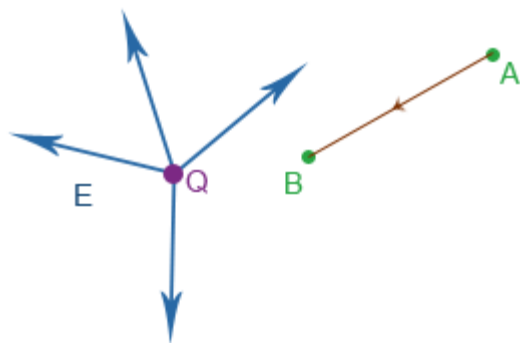
dr = infinitesimal vector displacement along the path.

- Potential energy can be defined only for conservative fields.

Electric Potential Energy

Work done by an external force to move a charge (against the electric force) from one point to another

- Compare with a spring: Do work in pushing it in, this work is stored as mechanical potential energy of spring.
- Similarly, push two like charges together, working against the electrical force, increasing its energy. This work is stored as electrical potential energy.



Work done by the external force

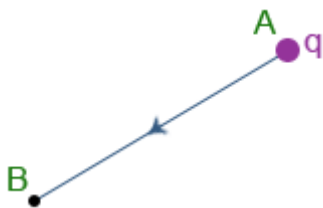
$$W_{AB} = U_B - U_A$$

Potential at B

Potential at A

Potential Difference

Work done by an external force in bringing a unit charge from one point to another (against the electric force)



$$W_{AB} = \frac{U_B - U_A}{q}$$

Infinity-potential is zero

For practical; earth –zero potential

Unit of potential difference,

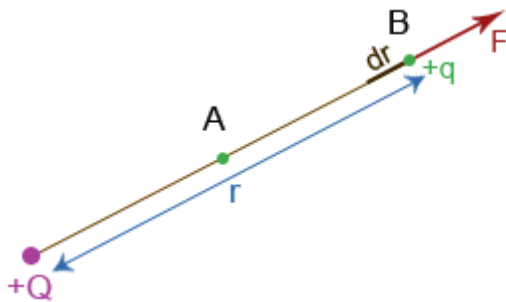
Thinking Question:

When you rub a balloon on your hair, the balloon can get charged to about -5000 V. The charge on it is less than a millionth of a Coulomb. Should you be worried about touching it?

A) yes

B) no

Potential at a Point



$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$$

Work done in bringing +q a distance $dr = dw$

$$dW = \underline{F} \cdot \underline{dr} = F \cos \pi dr = -F dr$$

$$dW = -\frac{Qq}{4\pi\epsilon_0} \left\{ \frac{1}{r^2} \right\} dr$$

$$W = -\int_A^B \frac{Qq}{4\pi\epsilon_0} \left\{ \frac{1}{r^2} \right\} dr$$

$$W = -\frac{Qq}{4\pi\epsilon_0} \int \left[\frac{1}{r^2} \right]_A^B dr \quad \text{Work done in bringing } +q \text{ from } A \rightarrow B$$

$$W = -\frac{Qq}{4\pi\epsilon_0} \left[-\left\{ \frac{1}{r_B} - \frac{1}{r_A} \right\} \right]$$

$$W = \frac{Qq}{4\pi\epsilon_0} \left\{ \frac{1}{r_B} - \frac{1}{r_A} \right\}$$

$$r_A \rightarrow \infty \Rightarrow \frac{1}{r_A} \rightarrow 0 \quad \text{Potential at infinity} = 0$$

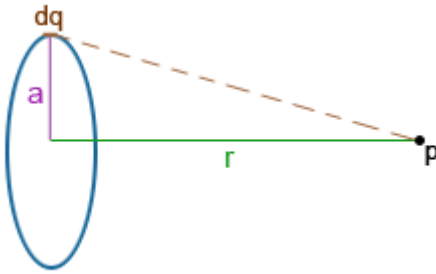
$$\Rightarrow W = \frac{Qq}{4\pi\epsilon_0} \frac{1}{r_B} \quad \text{This is for a } +q \text{ charge}$$

$\therefore$ For a unit positive charge i.e. $+q = 1$

$$W = V_B = \frac{Q}{4\pi\epsilon_0} \left\{ \frac{1}{r_B} \right\} \quad \text{Electric potential at B}$$

Calculation of electric Potential

1. Potential at a point P at a distance from a uniformly charged ring of radius a. (Charge Q)



Potential at P due to $dq = dv$

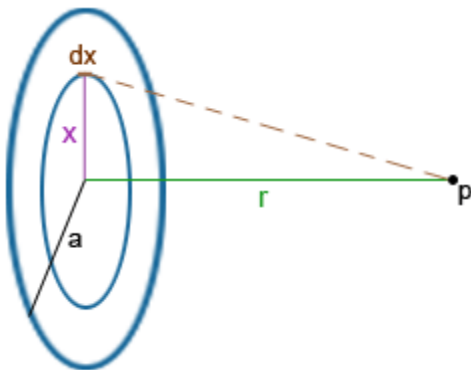
$$dq = dv = \frac{1}{4\pi\epsilon_0} \frac{dq}{(a^2 + r^2)^{1/2}}$$

$$\therefore \text{Potential at P due to the total ring} = \int dv = \frac{1}{4\pi\epsilon_0} \frac{1}{(a^2 + r^2)^{1/2}} \int dq$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{(a^2 + r^2)^{1/2}} \because \int dq = Q$$

2. Uniformly charged circular disc.

σ - Surface charge and a- radius

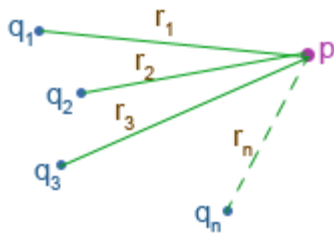


(Potential at P due to small ring) $dV = \frac{1}{4\pi\epsilon_0} \frac{2\pi x dx \sigma}{(r^2 + x^2)^{3/2}}$

Potential at P due to the total disc

$$\begin{aligned}
 V &= \int_0^a dV \\
 &= \frac{2\pi\sigma}{4\pi\epsilon_0} \int_0^a \frac{x dx}{(r^2 + x^2)^{3/2}} \\
 &= \frac{\sigma}{2\epsilon_0} [(r^2 + x^2)^{1/2}]_0^a \\
 &= \frac{\sigma}{2\epsilon_0} [(r^2 + a^2)^{1/2} - r]
 \end{aligned}$$

3. Several Point Charges



$$V_p = [V_1 + V_2 + V_3 + \dots + V_n]$$

$$V_p = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots + \frac{q_n}{r_n} \right\}$$

$$V_p = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

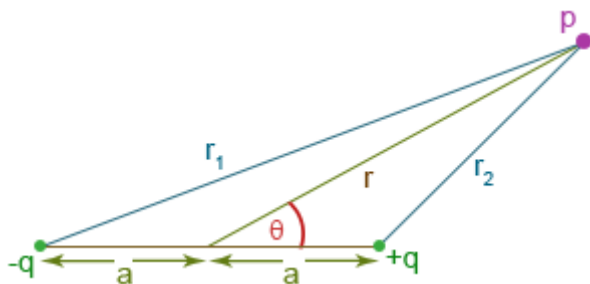
Potential is a scalar Quantity (not a vector quantity)

4. Potential due to an electric dipole

$$V_p = V_{+q} + V_{-q}$$

$$= \frac{q}{4\pi\epsilon_0 r_2} + \frac{(-q)}{4\pi\epsilon_0 r_1} = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{r_2} - \frac{1}{r_1} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \left(\frac{r_1 - r_2}{r_1 r_2} \right)$$



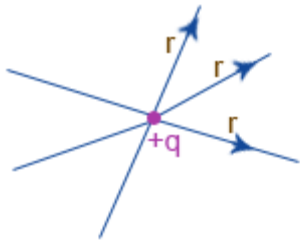
If $r_1, r_2 \gg a$

$$2a \cos \theta \cong r_1 - r_2; r_1 r_2 = r^2$$

$$\Rightarrow V_p = \frac{q}{4\pi\epsilon_0} \left\{ \frac{2a \cos \theta}{r^2} \right\} = \frac{P \cos \theta}{4\pi\epsilon_0 r^2}$$

$\therefore P = 2aq$ (Dipole moment)

Equipotential Surfaces



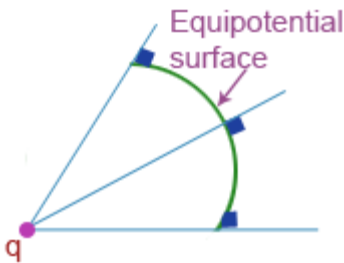
Consider a point charge, then the E is radial.

Where r is constant V is the same.

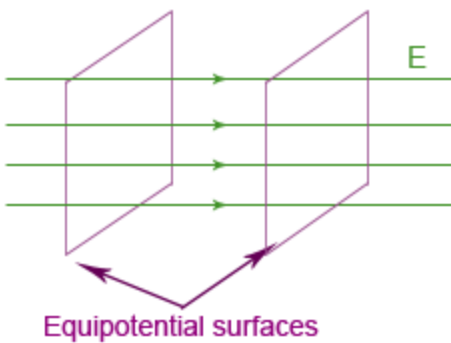
- Equipotential surface-in this case concentric sphere!

Definition:- The locus of the points of equal potential is called a equipotential surface.

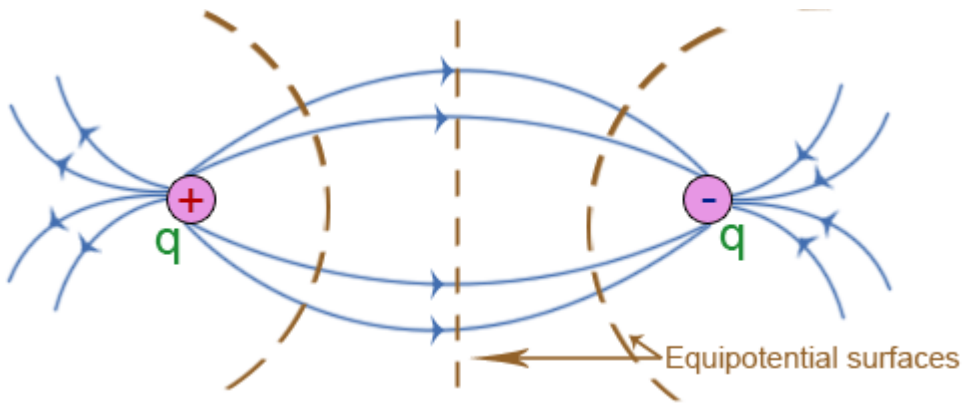
- Equipotential surfaces are perpendicular to the electric field.



For a uniform electric field of E

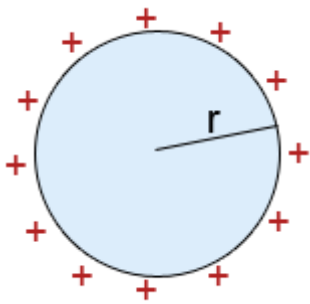


Two opposite charges of same magnitude

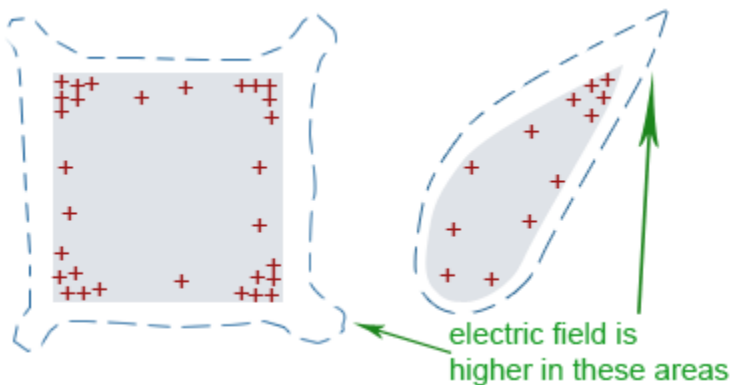


Charge distribution on conductors

- For a conductor charges are distributed only on the surface
- For a conducting sphere the charges are uniformly distributed.



Surface Charge density $\propto \frac{1}{R}$



- Electric field in these areas is extremely high
 - Air molecules are ionized – ionized molecules + free electrons
- Free electrons are accelerated in the electric field- ionize more gas molecules by collision –more free electrons –cumulative process
- ionized air molecules are violently repelled from the edge

This process is called point discharge or Corona discharge

- neutral minute particles touching the pointed surface, get charged (by contact) and repel violently from the edge---ELECTRIC WIND

E.g. Lightning conductors

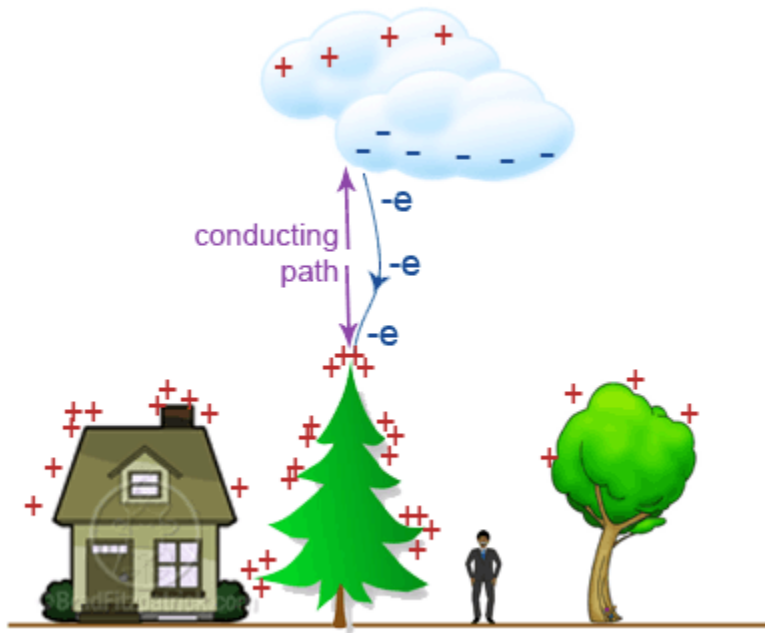


During a thunderstorm clouds are charged. Generally the positive charges are on the top of the cloud and negative charges are on the bottom. This implies negative charge will induce positive charge on earth (buildings /trees/ people....). That is very high potential difference is created ($\sim 5 \times 10^7 \text{V}$). The air suddenly gets ionized and discharge occurs damaging humans / buildings etc.

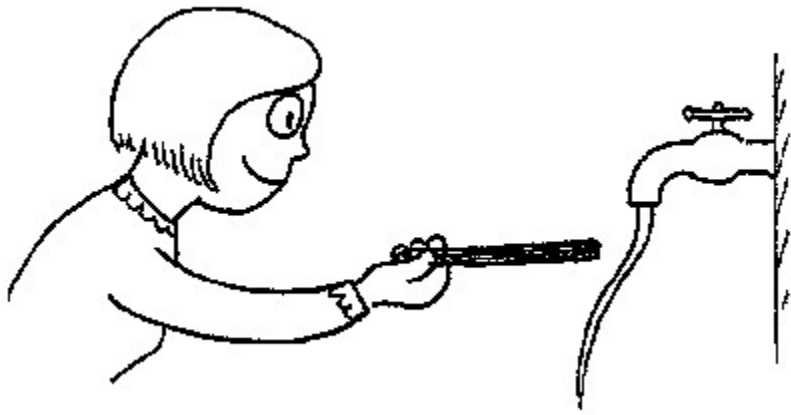
- to avoid this lightning conductors are used!

Lightning conductors have sharp edges

- They are highly charged due to inductance $\rightarrow$ corona discharge takes place $\rightarrow$ conducting path is created through the lightning conductors.



Thinking Question

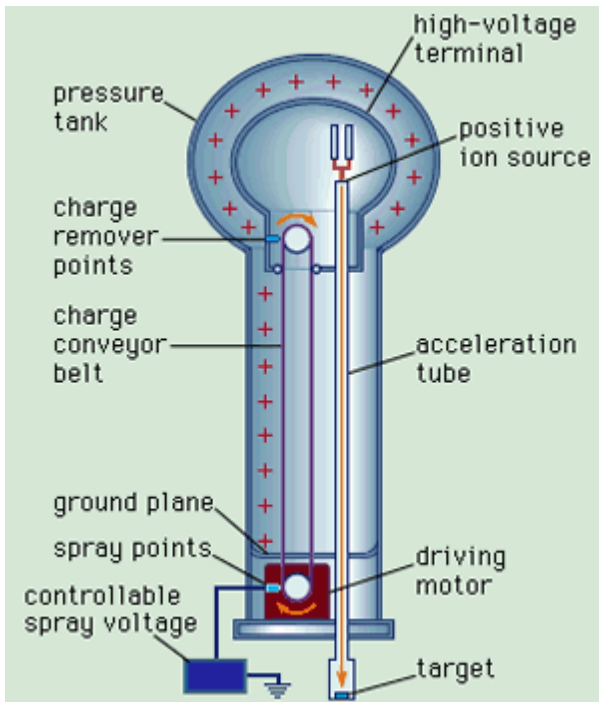


A thin stream of water bends toward a negatively charged rod. When a positively charged rod is placed near the stream, it will bend in the

1. Positive direction
2. Same direction
3. ... but it won't bend at all

Application of Corona discharge

1. Van de graff generator: a common device for building up high voltages



Needles maintained at large negative potential w.r.t. ground. They discharge electrons continuously onto the rubber belt which then carry them up into hollow conductor. Electrons end up on the outer sphere because there has to be 0 E-field inside → picked up by metal points (acting like lightning rods). Inside remains uncharged so more electrons keep coming up → end up with huge voltage on the dome. Can get as high as 20 million volts!

Can raise your hair with this !! Charges go into your hair, causing hairs to repel each other.

2. Discharging a charged body without touching.

Read!

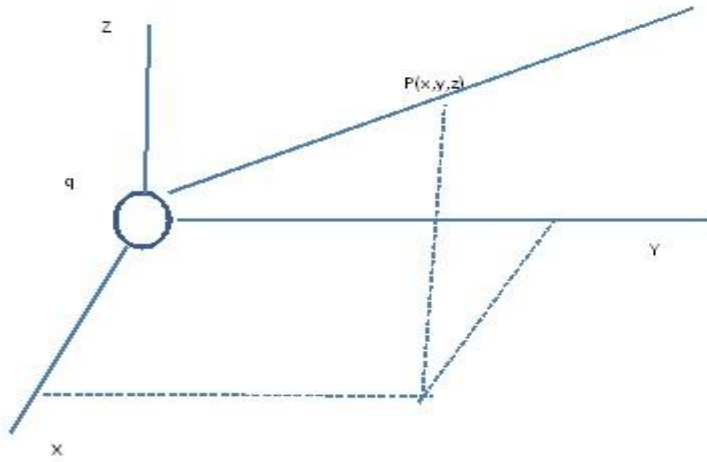
Thinking Question

When a power line falls near you, to move away should you

- A) Walk away from it?
- B) Hop with both feet together?
- C) Get down on your hands and knees and crawl away?

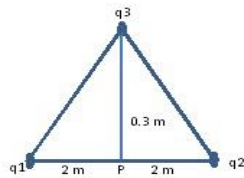
Exercise (Try them)

- (a) What is electric potential (often called as absolute electric potential) at a point in space and how is this related to the electric field?
(b) if $q = 40 \mu\text{C}$, $x = 0.5 \text{ m}$, $y = 0.8 \text{ m}$, $z = 0.6 \text{ m}$, what is the potential at point P in the figure below?



- A hollow sphere has a radius of 10 cm and bears a charge of 5 nC. If the sphere is isolated, what is the potential a distance of 50 cm from the centre of the sphere?
- What is the potential at the point midway between charges of $+2 \mu\text{C}$ and $+5 \mu\text{C}$ which are 6 m apart.
- Figure shows a triangular array of three point charges. Find the electric potential V of these source charges at the midpoint P of the ring. ($q_1 = 1 \times 10^{-6} \text{ C}$, $q_2 = -2 \times 10^{-6} \text{ C}$, $q_3 = 3 \times 10^{-6} \text{ C}$)

1.



5. A metal sphere 30 cm in radius is positively charged with $2\text{ }\mu\text{C}$. Find the potential at the (a) center of the sphere
(b) on the sphere
(c) 1 m from the center of the sphere
6. Two metal spheres (radii a and b) are very far apart but are connected by a thin wire. Their combined charge is Q .
(a) What is the charge on each?
(b) What is their absolute potential?
7. In certain region of space, the electric field is directed in the $+y$ direction and as a Magnitude of 4000 V/m . what is the potential difference from the coordinate origin to the following points?
(a) $x=0, y=20\text{ cm}, z=0$
(b) $x=0, y=-30\text{ cm}, z=0$
(c) $x=0, y=0, z=15\text{ cm}$